LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to speed and velocity, vectors, wind velocity, vector addition and triangle of velocity.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to speed and velocity, vectors, wind velocity, vector addition and triangle of velocity.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Vector addition diagram drawing; Safety: Ensure sharp pencils are capped; Supervision: Instructor verifies scale accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the difference between speed and velocity?
(b) Define a vector quantity.
(c) How is wind velocity represented?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of position on earth • Scalar vs Vector quantities • Speed and velocity definitions • Wind velocity components • Vector addition principles	L & Ds	PPT, IB	15 min
(b)	• Triangle of velocity construction • Head-to-tail vector method • Resultant vector calculation • Wind correction angle determination • Ground speed derivation	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of velocity triangles • Effect of wind direction changes • Magnitude and direction analysis • Real-world flight scenario application • Error sources in vector addition	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wind velocity affects ground speed.
(b) Describe the steps to construct a triangle of velocity.
(c) Differentiate between true airspeed and ground speed.

12. Home Assignment and Follow Up	Prepare a brief note on the navigation components required for the next lesson (LP 05) regarding the triangle of velocities.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to the navigation components of the triangle of velocities i.e. airspeed, Mach nos., ground speed, track and drift. Introduction to the application of the triangle of velocities in navigation.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min